

Ariel Campêlo Viana Morais

Backend Developer & Artificial Intelligence Engineer

CONTACT

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EDUCATION

B.Sc. in Mathematics
University of São Paulo (USP)
2017 – 2023

- Thesis: Machine Learning in Cardiovascular Medicine (InCor/FMUSP)
- POSCOMP & Graduate Coursework at IME-USP (KBS - Knowledge-Based Systems, Algorithms, Graphs - Grade A)

SKILLS & STACK

- Python (FastAPI / Pydantic)
- Twilio API & Webhooks
- WhatsApp Business & Redis
- Symbolic AI / Ontologies
- UN/CEFACT Standards (e-CERT)
- OWL / RDF / SPARQL
- NLP (spaCy / GloVe / NER)
- PyTorch & Vision (GANs)
- Vue.js 3 (Composition / Pinia)
- Docker & Jenkins (CI/CD)
- Kubernetes
- MySQL & Linux (HPC)

LANGUAGES

- Portuguese (Native)
- English (Proficient)
- French (Intermediate)

PROFESSIONAL SUMMARY

B.Sc. in **Mathematics from the University of São Paulo (USP)** specializing in **Symbolic Artificial Intelligence (Ontologies)**, **NLP**, and **Backend Software Engineering**. AI Researcher at **ITS** and Technology Consultant at **FGV/IBRE**, leading AI system architecture for international electronic sanitary certificate exchanges (e-CERT/MAPA) and agribusiness platforms (CONSECANA-SP). Creator of an ecosystem of self-built intelligent web apps (PWAs) to optimize daily productivity and personal well-being.

PROFESSIONAL EXPERIENCE & KEY PROJECTS

Institute of Software Technology (ITS) / Pirus | **AI & Software Researcher** 2021 – Present

- **e-CERT — Electronic Sanitary Certificate Exchange (MAPA / UN/CEFACT / OECD):** Lead AI/NLP architect for the Brazilian Ministry of Agriculture (MAPA). Built data normalization APIs using **spaCy (NER/textcat with GloVe)** aligned with **UNECE** vocabulary standards. Operating live with the **European Union, China, Russia, South Korea, Australia, Chile, and the Netherlands**.
- **Salvia (Full-Stack AI Clinical Assistant & GraphRAG):** Real-time automated patient anamnesis and doctor-patient communication engine via **WhatsApp (Twilio API, Webhooks, Redis)** engineered at Pirus. SPA built with **Vue.js (JavaScript, WebSockets)** integrated with **Python (FastAPI)** microservices, vector search (ChromaDB), and neuro-symbolic medical ontologies (**SNOMED CT, ICD-10**).
- **CONSECANA-SP & FGV / IBRE (Consultant 2024–2025):** Architecture of the Sugarcane Industry Indicator Platform with data harmonization under the SAT standard.
- **Port Community Systems (PCS / BRAEXP):** ICT architecture specification for the Santos Port Authority in partnership with ITS/USP (Scrum).

Heart Institute (HC - FMUSP) | **Bioinformatics & Big Data** 2018 – 2020

- **Cardiovascular Machine Learning:** Statistical predictive modeling (PLS/RCMQP in **R**) for genomic data (+300k individuals) under Prof. Dr. José Eduardo Krieger.

PRODUCTIVITY ECOSYSTEM, COMPUTER VISION & PROJECTS

🚀 **Personal Productivity & Well-Being Ecosystem** | **PWAs**

Conception and engineering of self-built tools designed to guide daily productivity, habit tracking, and physical well-being:

- 🧘 **MindFlow (Journaling & Habit Tracker):** Mobile-first PWA featuring a Tinder-style Swipe Game mode, 1-touch micro-journaling, and behavioral analytics.
- 🏋️ **Motion App (Calisthenics & Semantic Workouts):** Core engine in **FastAPI + RDFLib (OWL/SPARQL)** with camera pose estimation via **Vue 3 + TensorFlow.js (MoveNet)**.
- 🖼️ **Coloriz.AI | Computer Vision & Deep Learning**
- Historical media chromatic restoration for 1920s archives using **PyTorch (GANs / Autoencoders in CIE Lab space)**.